www.tntextbooks.in

groups such as CH_3O^-, $C_2H_5O^-$, COO^- etc

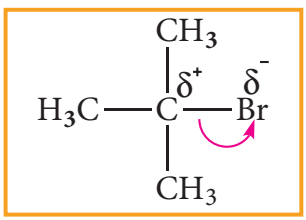
Lesser the electronegativity of the elements, greater is the +I effect. The relative order of +I effect of some alkyl groups is given below

$$-C(CH_3)_3 > -CH(CH_3)_2 > -CH_2CH_3 > -CH_3$$

Let us understand the influence of inductive effect on some properties of organic compounds.

Reactivity:

When a highly electronegative atom such as halogen is attached to a carbon then it makes the C-X bond polar. In such cases the -I effect of halogen facilitates the attack of an incoming nucleophile at the polarised carbon, and hence increases the reactivity.

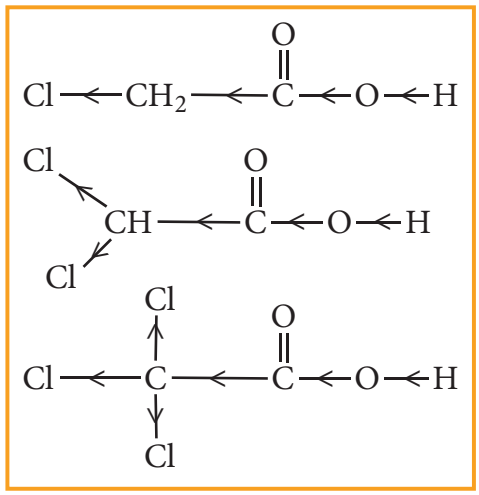


If a -I group is attached nearer to a carbonyl carbon, it decreases the availability of electron density on the carbonyl carbon, and hence increases the rate of the nucelophilic addition reaction.

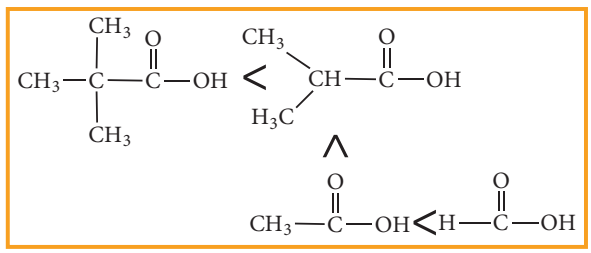
Acidity of carboxylic acids:

When a halogen atom is attached to the carbon which is nearer to the carboxylic acid group, its -I effect withdraws the bonded electrons towards itself and makes the ionisation of H^+ easy. The acidity of various chloro acetic acid is in the following order. The strength of the acid increases with increase in the -I effect of the group attached to the carboxyl group.

Trichloro acetic acid > Dichloro acetic acid > Chloro acetic acid > acetic acid



Similarly, the following order of acidity in the carboxylic acids is due to the +I effect of alkyl group.



Electrometric effect (E)

Electromeric is a temporary effect which operates in unsaturated compounds (containing >C=C<, >C=O, etc...) in the presence of an attacking reagent.

Let us consider two different compounds (i) compounds containing carbonyl group (>C=O) and (ii) unsaturated compounds such as alkenes (>C=C<).

When a nucleophile approaches the carbonyl compound, the π electrons between C and O is instantaneously shifted to the more electronegative oxygen. This makes the carbon electron deficient and thus facilitating the formation of a new bond